

VALOX™ Resin DR51 - Asia

Polybutylene Terephthalate

SABIC

PROSPECTOR®

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Technical Data

Product Description

15% GR polyester, excellent mechanical, thermal and electrical performance. Non-flame retardant. Spotlights, appliance housings, handles, connectors

General

Material Status	• Commercial: Active		
UL Yellow Card ¹	• E45587-236794 • E207780-100164507		
Search for UL Yellow Card	• SABIC • VALOX™ Resin		
Availability	• Asia Pacific		
Uses	• Aerospace Applications • Appliances • Automotive Exterior Parts • Automotive Interior Parts • Automotive Under the Hood • Construction Applications	• Electrical Parts • Electrical/Electronic Applications • Electronic Displays • Industrial Applications • Lawn and Garden Equipment • Lighting Applications	• Material Handling • Medical/Healthcare Applications • Outdoor Applications • Recreational Vehicle Applications • Water Management
Also Available In	• Europe	• Latin America	• North America

Physical	Nominal Value Unit	Test Method
Density / Specific Gravity	1.41 g/cm ³	ASTM D792
Specific Volume	0.710 cm ³ /g	ASTM D792
Molding Shrinkage		Internal Method
Across Flow : 1.50 to 3.20 mm	0.50 to 0.80 %	
Flow : 1.50 to 3.20 mm	0.40 to 0.60 %	
Across Flow : 3.20 to 4.60 mm	0.80 to 1.1 %	
Flow : 3.20 to 4.60 mm	0.60 to 0.90 %	
Water Absorption (24 hr, 23°C)	0.070 %	ASTM D570
Mechanical	Nominal Value Unit	Test Method
Tensile Strength ³ (Break)	93.0 MPa	ASTM D638
Tensile Elongation ³ (Break)	5.0 %	ASTM D638
Flexural Modulus ⁴ (50.0 mm Span)	4800 MPa	ASTM D790
Flexural Strength ⁴ (Break, 50.0 mm Span)	144 MPa	ASTM D790
Impact	Nominal Value Unit	Test Method
Notched Izod Impact (23°C)	53 J/m	ASTM D256
Unnotched Izod Impact (23°C)	530 J/m	ASTM D4812
Hardness	Nominal Value Unit	Test Method
Rockwell Hardness (R-Scale)	118	ASTM D785
Thermal	Nominal Value Unit	Test Method
Deflection Temperature Under Load		ASTM D648
0.45 MPa, Unannealed, 6.40 mm	210 °C	
1.8 MPa, Unannealed, 6.40 mm	190 °C	
CLTE - Flow		ASTM E831
-40 to 40°C	2.2E-5 cm/cm/°C	
60 to 138°C	2.2E-5 cm/cm/°C	
RTI Elec	130 °C	UL 746B
RTI Imp	130 °C	UL 746B
RTI Str	140 °C	UL 746B
Electrical	Nominal Value Unit	Test Method
Volume Resistivity	3.3E+16 ohms·cm	ASTM D257



Electrical	Nominal Value Unit	Test Method
Dielectric Strength		ASTM D149
1.60 mm, in Oil	23 kV/mm	
3.20 mm, in Air	20 kV/mm	
Dielectric Constant		ASTM D150
100 Hz	3.60	
1 MHz	3.40	
Dissipation Factor		ASTM D150
100 Hz	2.0E-3	
1 MHz	0.020	
Arc Resistance ⁵	PLC 5	ASTM D495
Comparative Tracking Index (CTI)	PLC 2	UL 746A
High Voltage Arc Resistance to Ignition (HVAR)	PLC 1	UL 746A

Flammability	Nominal Value Unit	Test Method
Flame Rating (0.71 mm)	HB	UL 94

Injection	Nominal Value Unit
Drying Temperature	120 °C
Drying Time	3.0 to 4.0 hr
Suggested Max Moisture	0.020 %
Suggested Shot Size	40 to 80 %
Rear Temperature	240 to 255 °C
Middle Temperature	245 to 260 °C
Front Temperature	250 to 265 °C
Nozzle Temperature	245 to 260 °C
Processing (Melt) Temp	250 to 265 °C
Mold Temperature	65 to 90 °C
Back Pressure	0.300 to 0.700 MPa
Screw Speed	50 to 80 rpm
Vent Depth	0.025 to 0.038 mm

Injection Notes

- Drying Time (Cumulative): 12 hr

Notes

¹ A UL Yellow Card contains UL-verified flammability and electrical characteristics. UL Prospector continually works to link Yellow Cards to individual plastic materials in Prospector, however this list may not include all of the appropriate links. It is important that you verify the association between these Yellow Cards and the plastic material found in Prospector. For a complete listing of Yellow Cards, visit the UL Yellow Card Search.

² Typical properties: these are not to be construed as specifications.

³ Type I, 5.0 mm/min

⁴ 1.3 mm/min

⁵ Tungsten Electrode



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Where to Buy

Supplier

SABIC

Web: <http://www.sabic.com/>

Distributor

3Polymer (Guangzhou) Chemical Technology Co., Ltd.

Telephone: +86-20-3466-7988

Web: <http://3polymer.com>

Availability: China

